SysTec Thermostat Architecture Recommendation

I recommend NXP's RW612 with external flash for SysTec's production thermostat. It combines the required peripheral interfaces and integrated Wi-Fi with more RAM than Microchip's WFI32E01PC. The Raspberry Pi 4B remains useful for developing and testing the Python prototype.

The prototype starts off at a 72°F set point. GPIO Zero button events cycle off, heat, and cool or change the set point by 1°F. I2C reads the AHT20; Fahrenheit readings are floored to whole degrees. Red pulses below the set point in heat mode; blue pulses above it in cool mode. Otherwise, the selected LED is solid; off extinguishes both. The LCD shows date/time and alternates temperature with mode/set point. UART sends state, temperature, and set point every 30 seconds at 115200 baud, 8N1.

Raspberry Pi 4B uses a 64-bit Cortex-A72 processor and Linux. A 2 GB model with a proposed 16 GB microSD card provides ample RAM and flash storage for Python, libraries, and logs; application storage is on microSD, not MCU program flash. Its header supports the existing I2C sensor, GPIO LCD and buttons, PWM LED control, and UART. Integrated dual-band 802.11ac supports cloud access. Retaining Python reduces migration work, although Linux adds scheduling variability and operating-system maintenance (Raspberry Pi Ltd., n.d.).

Microchip's WFI32E01PC integrates a 200 MHz MIPS MCU, 1 MB program flash, 256 KB RAM, and 2.4 GHz 802.11b/g/n Wi-Fi. Its I2C, UART, GPIO interrupts, and timer/output-compare resources support the sensor, serial link, three buttons, six LCD signals, and two fading LEDs. This compact module can run a C implementation with Microchip's networking stack. Its smaller RAM budget demands careful allocation for TLS buffers, stacks, and telemetry (Microchip Technology Inc., 2025).

NXP's RW612, representing the Freescale/NXP option, provides a 260 MHz Cortex-M33, 1.2 MB SRAM, dual-band Wi-Fi 6, and an external flash interface. Configurable I2C/UART interfaces, GPIO interrupts, and PWM timers support the same peripherals after pin mapping. I propose 8 MB external flash for firmware and update storage. C with MCUXpresso and FreeRTOS would replace the Linux-specific Python libraries. More RAM and hardware security favor future secure connectivity, at the cost of porting and board integration (NXP Semiconductors, n.d.).

For all three candidates, I would send telemetry through authenticated MQTT over TLS using their Wi-Fi interfaces, while local control continues during outages. The Pi has generous capacity; both MCU designs need measured firmware size and peak RAM usage. Before production approval, I would verify networking, update-image space, interrupt responsiveness, and at least 25% memory headroom under worst-case operation. These are acceptance targets, not measured results.

References

Microchip Technology Inc. (2025). *PIC32MZ W1 and WFI32 family* [Data sheet]. https://www.microchip.com/bin/mchp/product-ds.WFI32E01PC.pdf

NXP Semiconductors. (n.d.). *RW612*. https://www.nxp.com/products/RW612

Raspberry Pi Ltd. (n.d.). *Raspberry Pi 4 tech specs*. https://www.raspberrypi.com/products/raspberry-pi-4-model-b/specifications/